

Ruidong Li

PhD candidate, Department of Civil and Environmental Engineering, The Hong Kong Polytechnic University, Hung Hom, Kowloon, Hong Kong, China
Mobile: +(86)15107367771, +(852)93284617; Email: rui-dong.li@connect.polyu.hk
DOB: 16 July 1999

EDUCATION

2022-2026	THE HONG KONG POLYTECHNIC UNIVERSITY, HONG KONG Ph.D. in Geotechnical Engineering Title: AI-Enhanced Granular Mechanics: From μ CT-Based Triaxial Tests to GPU-Accelerated Digital Twins. Supervisor: Prof. Zhen-Yu Yin.
2021-2022	THE HONG KONG POLYTECHNIC UNIVERSITY, HONG KONG M.S. in Civil Engineering Distinction, awarded the Best Dissertation.
2017-2021	CENTRAL SOUTH UNIVERSITY, CHANGSHA B.E. in Civil Engineering First Class Honours Equivalent, overall grade: 91.1%, ranked 2/90.

PROFESSIONAL EXPERIENCES

2023-2023	UNIVERSITY OF CAMBRIDGE, CAMBRIDGE Visiting Scholar Collaborated on my PhD research on AI-enhanced 3D particle reconstruction. Host supervisor: Prof. Brian Sheil.
2023-2023	UNIVERSITY OF CAMBRIDGE, CAMBRIDGE MCR Social Membership Offered by Downing College.
2021-2022	THE HONG KONG POLYTECHNIC UNIVERSITY, HONG KONG Research Assistant Worked at the Department of Land Surveying and Geo-Informatics for developing intelligent traffic systems using computer vision. Host supervisor: Prof. Tony Chengxiang Zhuge.

AWARDS & HONORS

<u>Year</u>	<u>Title</u>	<u>Organization</u>
2025	Best Poster Award	The 10th International Conference on Discrete Element Methods (DEM10)
2022	Best MSC Dissertation Award	The Hong Kong Polytechnic University
2021	Outstanding Graduates	Hunan Province
2021	Outstanding Graduates	Central South University
2020	China Railway First Class Scholarship	China Railway Group Limited
2020	First Prize of the 11th National BIM-CIM Competition	China Construction Education Association
2020	First Class Scholarship	Central South University
2019	National Scholarship	Ministry of Education of the People's Republic of China
2019	First Class Scholarship	Central South University
2018	First Class Scholarship	Central South University
2017	Second Class Scholarship	Central South University

GRANTS & FUNDING

Total: HK\$2.1M

<u>Year</u>	<u>FEC Value</u>	<u>Title</u>
2025	CNY50k (Co-I)	Deep learning representation of hot dry rock cracks and analysis of crack propagation mechanism under multi-field coupling conditions. Funding body: The State Key Laboratory of Intelligent Construction and Healthy Operation and Maintenance of Deep Underground Engineering of China
2024	HK\$1.1m (Co-I)	Machine Learning-Aided Multiscale Approach for Recycled Solid Waste Mixtures Composed of Easily Crushable Rigid Aggregates and Highly Deformable Soft Aggregates. Funding body: Research Grants Council of Hong Kong
2022	HK\$911k (Co-I)	Deep Learning-based Multiscale Approach for Granular Materials from Particle Recognition to Mechanical Modelling. Funding body: Research Grants Council of Hong Kong

SERVICES

Journal Reviewer

Computers and Geotechnics (Elsevier), Granular Matter (Springer), Engineering with Computers (Springer), Proceedings of the Institution of Civil Engineers - Structures and Buildings (Emerald), Scientific Reports (Springer), Applied Mathematics in Science and Engineering (Taylor&Francis), Discover Applied Sciences (Springer)

SUPERVISION

Undergraduate Student

2025-2026: Lui Chi Hung (Final Year Project), The Hong Kong Polytechnic University (co-supervisor)

2025-2026: Wang Chenyi (Final Year Project), The Hong Kong Polytechnic University (co-supervisor)

PUBLICATIONS

(In total: **15**, first/corresponding: **7**; *corresponding; updated in **July, 2026**)

Google Scholar: <https://scholar.google.com/citations?user=FwgutEcAAAAJ&hl=zh-CN> (Citations: 264, h-index: 8)

Scopus: <https://www.scopus.com/authid/detail.uri?origin=resultslist&authorId=57223237262&zone=> (Citations: 235, h-index: 7)

Web of Science: <https://www.webofscience.com/wos/author/record/OXB-5529-2025> (Citations: 206, h-index: 7)

2026

1. He SH, Yin ZY*, Ding Z, **Li RD**. (2026) Development of a novel high-pressure triaxial apparatus and investigation of the undrained behavior of sand in ultra-deep sea environment. *Canadian Geotechnical Journal*. DOI: 10.1139/cgj-2026-0066.
2. He SH, Yin ZY*, Ding Z, **Li RD**, Liao D. (2026) Drained deformation and non-coaxiality of granular materials under pure principal stress rotation: role of particle morphology. *Acta Geotechnica*. DOI: 10.1007/s11440-025-02640-9.
3. **Li RD**, Jiang HR*, Sheil B, Xu CK, He SH*. (2026) Effects of particle morphology and intermediate principal stress ratio on the multiscale granular behavior: Insights from μ CT-informed LS-DEM modeling. *Canadian Geotechnical Journal*. DOI: 10.1139/cgj-2025-0949.
4. **Li RD**, Yin ZY*, He SH. (2026) Physics-Informed Spatiotemporal Optimal Transport for Tracking Particle and Breakage. *Journal of Geotechnical and Geoenvironmental Engineering*. DOI: 10.1061/JGGEFK.GTENG-14484.
5. **Li RD**, Yin ZY*, He SH, Wu MM. (2026) Novel particle reconstruction and tracking algorithms to reveal 3D micromechanical behaviors of coral sands. *Canadian Geotechnical Journal*. DOI: 10.1139/cgj-2025-0655.

2025

6. He SH, Yin ZY*, Ding Z, **Li RD**. (2025) Strength and Deformation Characteristics of Sand–Rubber Mixtures under Torsional Shear Loadings. *Journal of Geotechnical and Geoenvironmental Engineering*. DOI:

10.1061/JGGEFK.GTENG-13008.

7. Zeng KF, Yin ZY*, **Li RD**, Jin YF. (2025) Role of particle morphology in monotonic and cyclic behavior of granular materials: Insights from cereals. *Powder Technology*. DOI: 10.1016/j.powtec.2025.121452.
8. Jiang HR, Debanath OC*, Li Y, **Li RD**. (2025) A level-set method-based framework for modeling abrasion of railway ballast. *Transportation Geotechnics*. DOI: 10.1016/j.trgeo.2025.101628.
9. He SH, Yin ZY*, Ding Z, **Li RD**. (2025) Novel observations for the impact of particle morphology on shear modulus of granular materials. *Acta Geotechnica*. DOI: 10.1007/s11440-025-02640-9.
10. **Li RD**, Yin ZY*, He SH, Sheil B. (2025) Enhancement and assessment of large vision models for 3D particle reconstruction from X-ray tomography. *Canadian Geotechnical Journal*. DOI: 10.1139/cgj-2024-0736.
11. **Li RD**, Yin ZY*, He SH. (2025) 3D reconstruction of arbitrary granular media utilizing vision foundation model. *Applied Soft Computing*. DOI: 10.1016/j.asoc.2024.112599.

2024

12. He SH, Yin ZY*, Ding Z, **Li RD**. (2024) Particle morphology and principal stress direction dependent strength anisotropy through torsional shear testing. *Canadian Geotechnical Journal*. DOI: 10.1139/cgj-2023-0717.
13. **Li RD**, Zhang P*, Yin ZY*, Sheil B. (2024) Enhanced Hybrid Algorithms for Segmentation and Reconstruction of Granular Grains from X-ray Micro Computed-Tomography Images. *International Journal of Numerical and Analytical Methods in Geomechanics*. DOI: 10.1002/nag.3832. (☆**Top 10 most-cited & most-read articles in 2024**)

2022

14. **Li RD**, Hu XY, Chen F, Wang X*, Xiong H, Wu H. (2022) A systematic framework for DEM study of realistic gravel-sand mixture from particle recognition to macro-and micro-mechanical analysis. *Transportation Geotechnics*. DOI: 10.1016/j.trgeo.2021.100693.

2021

15. Yang NY, Chen XB*, **Li RD**, Zhang JQ, Hu H, Zhang JS. (2021) Mesoscale numerical investigation of the effects of fiber stiffness on the shear behavior of fiber-reinforced granular soil. *Computers and Geotechnics*. DOI: 10.1016/j.compgeo.2021.104259.

Under Review

1. **Li RD**, Yin ZY*, Xu CK, He SH, Yang J*. An automated AI-augmented digital twin framework for multiscale modeling of granular materials. (submitted to *International Journal of Mechanical Sciences*, Major Revision).
2. **Li RD**, Yin ZY*, He SH, Zeng KF. Influence of Rubber Particle Shape on Rubber–Coral Sand Mixtures: A μ CT-Based Triaxial Study. (submitted to *Journal of Geotechnical and Geoenvironmental Engineering*).
3. **Li RD**, Yin ZY*. Micromechanics of crushable carbonate sand revealed by in situ X-ray microtomography. (submitted to *Géotechnique*).
4. **Li RD**, Yin ZY*, Zeng KF, He SH. Multiscale response of coral sand–rubber mixtures with different rubber contents revealed by X-ray microtomography. (submitted to *Canadian Geotechnical Journal*).
5. **Li RD**, Jiang HR*, Wang CY, Zeng KF. Novel framework for 3D reconstruction of multiphase granular mixtures from μ CT images. (submitted to *Computers and Geotechnics*).

Ready to Submit

1. **Li RD**, Yin ZY*. 3D in-situ micromechanical insights into the morphology-dependent behaviour of coral sands.
2. **Li RD**, Yin ZY*. Micromechanical insights into geomaterials by in-situ μ CT-based triaxial tests.
3. **Li RD**, Yin ZY*, Shi YH, Yang DF. Novel GPU-accelerated digital twin modeling framework for granular media.